

MBA Fastrack 2025 (CAT + OMETs)

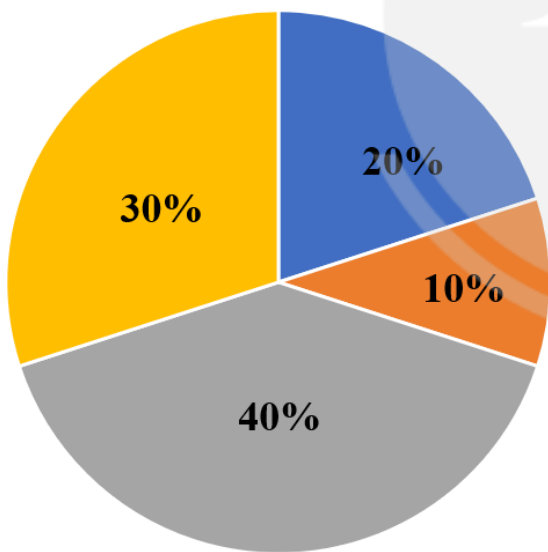
Data Interpretation & Logical Reasoning

DPP: 2

Pie Charts, Word Problems

Directions (1-5) Read the following passage and answer the given questions.

An ice-cream-making company has four outlets (A1, A2, A3, and A4) in a city and only four types of ice creams (Vanilla, Strawberry, Chocolate, and Butterscotch) are sold in the four outlets. Each outlet sells every kind of ice cream. After analysis of the sales (by value) made by each outlet in the last year, the company found that the following pie diagram represents the distribution of annual sales among four outlets. It also means the distribution of the sale of four ice creams in each of the four outlets and it also represents the distribution of sale of each ice cream across the four outlets.



Further, it is known that in the last one year:

- For any ice cream, the sale of the first outlet (A1) is more than the second outlet (A2) which in turn is more than the third outlet (A3) and which in turn is more than the fourth outlet (A4).
- In any outlet, the sale of the first ice cream (Vanilla) is more than the second ice cream

(Strawberry) which in turn is more than the third ice cream (Chocolate) and which in turn is more than the fourth ice cream (Butterscotch).

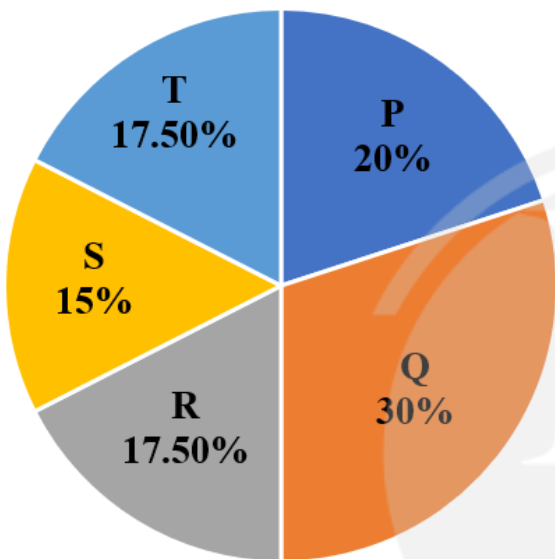
- Q1** What is the ratio of sales of chocolate ice cream in A3 to Butterscotch in A2?
 (A) 4:1 (B) 4:3
 (C) 3:1 (D) 1:1
- Q2** Which of the following statements is definitely TRUE for the sale of the last year in the four outlets?
 (A) Chocolate ice cream in A1 > Vanilla ice cream in A3 > Strawberry ice cream in A4
 (B) Butterscotch ice cream in A2 = Strawberry ice cream in A4 < Chocolate ice cream in A3
 (C) Vanilla ice cream in A1 > Strawberry ice cream in A2 = Chocolate ice cream in A3
 (D) None of these
- Q3** If A2 and A3 stop selling the Strawberry and Chocolate ice creams in the present year and everything else remains the same as last year, what will be the percentage reduction in total sales this year with respect to the previous year?
 (A) 25% (B) 20%
 (C) 30% (D) 33.33%
- Q4** What percentage of total revenue has been generated from the sale of the Strawberry ice cream in the A3 and Butterscotch ice cream in the A4 in the last year?
 (A) 10% (B) 12%
 (C) 8% (D) None of these



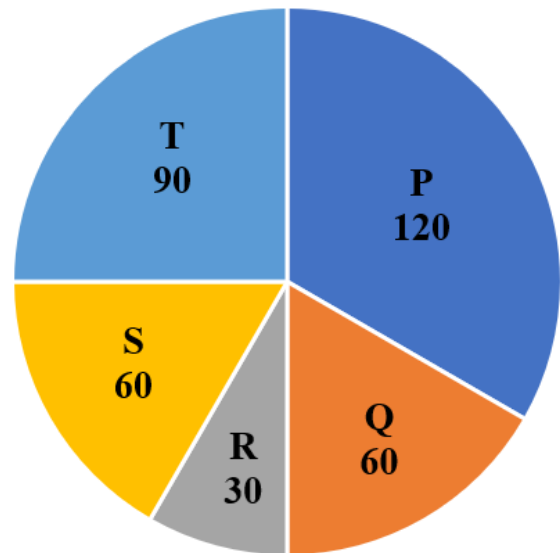
Q5 What is the minimum number of unique “outlet & ice cream” combinations that together generated exactly 50% of the revenue in the last year?

Directions (6-10) Read the following passage and answer the given questions.

The first Pie chart given below shows the percent distribution of the total number of students (Boys + Girls) in five different classes. The total number of students in all the classes together is 400.



The second pie chart shows the degree distribution of the difference between the number of boys and that of girls for all those five classes. The sum of differences between the number of boys and that of girls for all the classes is 120.



All the data is given for 2017.

Note: The number of boys is more than that of girls for classes P and R only.

Q6 What is the ratio of the total number of boys in classes P and T together to the total number of girls in classes P and Q together?
 (A) 6 : 7 (B) 7 : 8
 (C) 8 : 9 (D) 9 : 10

Q7 If in the year 2018, 60 new students took admission in class P after which the number of boys and the number of girls in that class in 2018 becomes same, then what is the ratio of number of new boys to that of new girls who took admission in class P in 2018? (All the other data remains the same.)
 (A) 3 : 8 (B) 2 : 7
 (C) 1 : 5 (D) 4 : 9

Q8 Ratio of number of boys to that of girls in class R in 2018 was 6 : 7 and some new students joined class R in 2018. If a total of 60 new students joined class R in 2018, then what is the difference between the number of new boys who joined class R in 2018 and the number of new girls who joined class R in 2018? (Assume all the other data remains the same).
 (A) 30 (B) 20
 (C) 25 (D) 15



Q9 If in other class U, the number of boys is 20% more than that in class T and the number of girls in class U is 14 less than that in class T, then the number of boys in class U is what percent of total number of girls in that class?

- (A) 72.67% (B) 78.33%
(C) 55.55% (D) 66.67%

Q10 Total number of boys in Class P is what percent more than the total number of girls in Class T? (only numerical value)

Directions (11-15) Read the following passage and answer the given questions.

Jatin wrote six exams, each in a different subject, among Maths, Physics, Chemistry, Economics, Botany, and Zoology. The pie chart, which is drawn to scale, provides the marks scored by Jatin in each subject as a percentage of the total marks obtained by him in all six subjects put together. The names of the subjects have intentionally been left out in the pie chart; instead, the subjects are labelled as X1 to X6, in no particular order.

Marks



■ X1 ■ X2 ■ X3 ■ X4 ■ X5 ■ X6

Further, it is known that:

1) In the pie chart, X1, X2 and X3 subtend angles 90° , 45° and 45° respectively.

2) Jatin scored 15 marks less in Chemistry than in Economics but he scored more in Chemistry as compared to Physics.

3) The angle that X4 subtends at the centre is smaller than that subtended by X3, while the angle that X6 subtends at the centre is larger than that subtended by X5.

4) The marks scored by Jatin in Maths are the same as the sum of the marks scored by him in two other subjects, in each of which Jatin scored an equal number of marks.

5) The difference between the highest marks that Jatin scored in any subject and the lowest marks that Jatin scored in any subject is 55.

6) Jatin scored 30 marks less in Zoology than in Chemistry.

7) The marks in Physics and Zoology are different.

Q11 What are the total marks scored by Jatin across the six subjects?

- (A) 280 (B) 320
(C) 360 (D) 400

Q12 What is the angle (in degree) subtended in the pie chart by the sector which represents the marks scored by Jatin in Chemistry?

- (A) 90 (B) 45
(C) 65 (D) 35

Q13 Which of the regions is represented by Zoology?

- (A) X1 (B) X2
(C) X5 (D) X4

Q14 How many marks did Jatin scored in Botany?

- (A) 45
(B) 35
(C) 65
(D) Either 35 or 65

Q15 In the given pie chart, the sector representing which of the following



subjects is definitely adjacent to the sector representing Zoology?

- (A) Economy (B) Botany
(C) Chemistry (D) Physics

Directions (16-20) Read the following passage and answer the given questions.

A startup company was established on January 1, 2021, with a total of ten members. These members were initially divided into four groups: P, Q, R, and S, with each group consisting of at least two members. Each member had a distinct integer age as of January 1, 2021, with the youngest being 25 years old and the oldest 34 years old. The group composition remained unchanged throughout each year.

At the end of each year, the members were reshuffled in such a way that no group had the same members as in the previous years. The total number of members in the company remained constant, meaning no one was added or removed over the years.

The table below shows the average ages of the members in each group on January 1st of three consecutive years:

Group	2021	2022	2023
P	25.5	29.33	34.5
Q	27.5	30.5	29.5
R	29.5	33	28
S	32.5	28.5	31

Additionally, it is known that Group P had four members on January 1, 2023.

Q16 What was the age of the eldest person in Group Q on January 1, 2021?

- (A) 32 years (B) 31 years
(C) 30 years (D) 28 years

Q17 How many members were common between Group P for the year 2021 and Group S for the year 2022?

- (A) 2 (B) 1
(C) 0 (D) 3

Q18

What was the total age of the members in Group Q on January 1, 2022?

- (A) 61 years (B) 70 years
(C) 122 years (D) None of these

Q19 Which group showed the highest increase in the sum of total age from 2021 to 2023?

- (A) Group P (B) Group Q
(C) Group R (D) Group S

Q20 What was the age of the youngest person in Group P on January 1, 2023?

[Type 'O' if the value cannot be uniquely determined]

Directions (21-25) Read the following passage and answer the given questions.

In a modified version of the Arts Proficiency Test (APT), three sections—Singing, Poetry, and History—are included, with 15 questions in each section. Four students from Rabindrakala Academy—Benny, Dayal, Prasoon, and Rekha—participated in the test.

Marking Scheme:

- If a student answers 1 to 5 questions correctly in a section, they receive 1 mark per correct answer.
- If a student answers 6 to 10 questions correctly, they receive 2 marks per correct answer.
- If a student answers more than 10 questions correctly, they receive 3 marks per correct answer.
- All 15 questions in each section must be attempted.
- There is no negative marking for incorrect answers.
- Additionally, if a student answers 2 or fewer questions incorrectly in a section, they receive 4 bonus marks for accuracy.

Given Information:

1. Benny answered 8 questions correctly in Singing, and his total score was 63.



2. Dayal answered 6 questions correctly in History and had a total of 21 correct answers across all sections.
3. Prasoon answered 7 questions correctly in Singing and 9 in History.
4. Rekha answered 7 questions correctly in Poetry, with a total of 18 correct answers.
5. The total number of correct answers in Singing across all four students is 34, and in Poetry, it is 36.

Note:

- No student answered the same number of questions correctly in any two sections.
- Every student answered at least 2 questions correctly in each section.
- Benny's correct answers in History are the same as Dayal's correct answers in Poetry.

- Q21** For how many of the four students, can the questions answered correctly by them be uniquely determined?
- Q22** What is the total number of questions that Benny answered correctly?
[Type '0' if the value cannot be uniquely determined]
- Q23** Which participant would have stood first in the APT exam?
(A) Benny
(B) Dayal
(C) Prasoon
(D) Cannot be determined
- Q24** What can be BEST said about Dayal?
(A) His total score lies between 50 to 55.
(B) He answered 6 questions correctly in Poetry.
(C) He answered the second-lowest number of correct questions in Poetry.
(D) He answered the highest number of correct questions in Singing among all four students.
- Q25** What can be BEST said about Prasoon?
- (A) He got 12 questions correct in Poetry.
(B) His total score is the second lowest.
(C) His total score is the arithmetic mean of the total scores of Dayal and Benny.
(D) His total score is the second highest.

Directions (26-30) Read the following passage and answer the given questions.

Biryani Hut sells only chicken biryani to its customers. In every chicken biryani 250 gm rice, 200 gm chicken, 1 boiled egg is there. In making 1 plate of chicken biryani spices of 30 Rupees is used. The cook's cost per day is 500 INR who can handle at most 10 biryanis a day. Beyond that requirement, one more cook is needed. Assume that other than the cook, there is no other employee cost. A dozen of egg is purchased at 60 INR and processing all the food items in a plate costs 60 INR additionally per plate. The monthly rent for the kitchen space is 15,000 INR. Price of rice and chicken per kg are 60 INR and 150 INR respectively. Fixed costs other than salary per month is 10,000 INR. Consider 30 days in a month on an average for this question. Also assume that other than the aforementioned cost points no other costs are there unless otherwise specified in the respective questions.

- Q26** If Biryani Hut sells each of the biryanis at 300 INR. Then minimum how many biryanis needed to be sold per month to make sure no losses are incurred. Assume that every day has the same number of plates of biryani demand.
(A) 250 (B) 260
(C) 270 (D) 280
- Q27** If Biryani Hut sells each of the biryanis at 240 INR. Then minimum how many plates of biryani needed to be sold per month to make sure no losses are incurred. Assume that every day has exactly the same number of plates of biryani demand.
(A) 550 (B) 560
(C) 570 (D) 580



Q28 Biryani Hut sells each of the biryanis at 240 INR. Then find out what can be the maximum possible profit or minimum possible loss if a total of 550 plates are sold.

- (A) 2500 (B) 2550
(C) 3000 (D) 3500

Q29 If the price of biryani dropped to 200 INR/plate how much profit/loss will be incurred by Biryani Hut to sell 900 plates of biryanis a month. Assume that the demand per day is the same.

- (A) 15,000 INR loss
(B) 16,000 INR loss

- (C) 17,000 INR loss
(D) 18,000 INR loss

Q30 Biryani Hut has optimised the processing and reduced the processing cost by 30 INR/plate. If the price of biryani dropped to 200 INR/plate how much profit/loss will be incurred by Biryani Hut to sell 900 plates of biryanis a month. Assume that the demand per day is the same.

- (A) 10,000 INR profit
(B) 11,000 INR profit
(C) 12,000 INR profit
(D) 13,000 INR profit



Answer Key

Q1 (B)
Q2 (B)
Q3 (A)
Q4 (D)
Q5 5
Q6 (C)
Q7 (C)
Q8 (B)
Q9 (D)
Q10 20
Q11 (C)
Q12 (C)
Q13 (D)
Q14 (A)
Q15 (C)

Q16 (D)
Q17 (B)
Q18 (A)
Q19 (A)
Q20 33
Q21 4
Q22 25
Q23 (C)
Q24 (D)
Q25 (A)
Q26 (C)
Q27 (C)
Q28 (A)
Q29 (B)
Q30 (B)



Hints & Solutions

Note: scan the QR code to watch video solution

Q1. Text Solution:

From the information given, we can conclude that:

Outlet	Share
A1	40%
A2	30%
A3	20%
A4	10%

Ice cream	Share
Vanilla	40%
Strawberry	30%
Chocolate	20%
Butterscotch	10%

Based on the above analysis, we create the following table showing sales figures in the four outlets:

	A1	A2	A3	A4	Total
Vanilla	16k	12k	8k	4k	40k
Strawberry	12k	9k	6k	3k	30k
Chocolate	8k	6k	4k	2k	20k
Butterscotch	4k	3k	2k	1k	10k
Total	40k	30k	20k	10k	100k

(Here, 'k' is a constant.)

Required ratio = $4k:3k = 4:3$.

Video Solution:



Q2. Text Solution:

	A1	A2	A3	A4	Total
Vanilla	16k	12k	8k	4k	40k
Strawberry	12k	9k	6k	3k	30k
Chocolate	8k	6k	4k	2k	20k

Butterscotch	4k	3k	2k	1k	10k
Total	40k	30k	20k	10k	100k

According to the first statement, $8k > 8k > 3k$ is False.

According to the second statement, $3k = 3k < 4k$ is True.

According to the third statement, $16k > 9k = 4k$ is False.

Video Solution:



Q3. Text Solution:

	A1	A2	A3	A4	Total
Vanilla	16k	12k	8k	4k	40k
Strawberry	12k	9k	6k	3k	30k
Chocolate	8k	6k	4k	2k	20k
Butterscotch	4k	3k	2k	1k	10k
Total	40k	30k	20k	10k	100k

Total sales in last one year = 100k

Reduction in sales in this year = $9k + 6k + 6k + 4k = 25k$

Thus, percentage reduction in sales = 25%

Video Solution:



Q4. Text Solution:

	A1	A2	A3	A4	Total
Vanilla	16k	12k	8k	4k	40k
Strawberry	12k	9k	6k	3k	30k



Chocolate	8k	6k	4k	2k	20k
Butterscotch	4k	3k	2k	k	10k
Total	40k	30k	20k	10k	100k

Sale of Strawberry ice cream in the A3 = 6k and

Butterscotch ice cream in the A4 = k

Total = 6k + k = 7k

Hence, the answer is 7%

Video Solution:



Q5. Text Solution:

	A1	A2	A3	A4	Total
Vanilla	16k	12k	8k	4k	40k
Strawberry	12k	9k	6k	3k	30k
Chocolate	8k	6k	4k	2k	20k
Butterscotch	4k	3k	2k	k	10k
Total	40k	30k	20k	10k	100k

To minimize the number of unique “outlet & ice cream” combinations that together generated exactly 50% of the revenue in the last year, we should include the largest sales figures.

Thus, $(16 + 12 + 12 + 9 + 1) = 50$ is a possible answer.

Hence, the answer is 5.

Video Solution:



Q6. Text Solution:

Classes	Total	Difference	Boys	Girls
P	20% of 400 = 80	$120 \times \frac{120}{360} = 40$	$\frac{80+40}{2} = 60$	$\frac{80-40}{2} = 20$

Q	30% of 400 = 120	$120 \times \frac{60}{360} = 20$	$\frac{120-20}{2} = 50$	$\frac{120+20}{2} = 70$
R	17.5% of 400 = 70	$120 \times \frac{30}{360} = 10$	$\frac{70+10}{2} = 40$	$\frac{70-10}{2} = 30$
S	15% of 400 = 60	$120 \times \frac{60}{360} = 20$	$\frac{60-20}{2} = 20$	$\frac{60+20}{2} = 40$
T	17.5% of 400 = 70	$120 \times \frac{90}{360} = 30$	$\frac{70-30}{2} = 20$	$\frac{70+30}{2} = 50$

Total number of boys in classes P and T together
= 60 + 20 = 80

Total number of girls in classes P and Q together
= 20 + 70 = 90

Required ratio = 80: 90 = **8: 9**

Video Solution:



Q7. Text Solution:

Classes	Total	Difference	Boys	Girls
P	80	40	60	20
Q	120	20	50	70
R	70	10	40	30
S	60	20	20	40
T	70	30	20	50

Let the number of new boys and the number of new girls who took admission in class P in 2018 be 'x' and '60 - x' respectively.

According to the question:

$$(60 + x) = 20 + (60 - x)$$

$$x = 20 - x$$

$$x = 10$$

Required ratio = x: (60 - x) = 10: 50 = **1: 5**.

Video Solution:



**Q8. Text Solution:**

Classes	Total	Difference	Boys	Girls
P	80	40	60	20
Q	120	20	50	70
R	70	10	40	30
S	60	20	20	40
T	70	30	20	50

Let the number of boys and that of girls in class R in 2018 be '6x' and '7x' respectively.

According to the question:

$$(6x - 40) + (7x - 30) = 60$$

$$13x = 130$$

$$x = 10$$

$$\text{Total new boys who joined class R in 2018} = (6x - 40) = 20$$

$$\text{Total new girls who joined class R in 2018} = (7x - 30) = 40$$

$$\text{Required difference} = 40 - 20 = 20$$

Video Solution:**Q9. Text Solution:**

Classes	Total	Difference	Boys	Girls
P	80	40	60	20
Q	120	20	50	70
R	70	10	40	30
S	60	20	20	40
T	70	30	20	50

$$\text{Total boys in class U} = 120\% \text{ of } 20 = 24$$

$$\text{Total girls in class U} = 50 - 14 = 36$$

$$\text{Required percent} = \frac{24}{36} \times 100 = 66.67\%$$

Video Solution:**Q10. Text Solution:**

Classes	Total	Difference	Boys	Girls
P	80	40	60	20
Q	120	20	50	70
R	70	10	40	30
S	60	20	20	40
T	70	30	20	50

$$\text{Total number of boys in Class P} = 60$$

$$\text{Total number of girls in Class T} = 50$$

$$\text{Required answer} = \frac{60-50}{50} \times 100 = 20\%$$

Video Solution:**Q11. Text Solution:**

From (iv), Jatin's score in Maths is the sum of the scores in two other subjects and he scored equally in the two subjects. The only case is X1, which can be expressed as the sum of two other pies, i.e. X2 and X3.

In this case, X2 and X3 are equal (from (i)).

Hence, X1 must be Maths.

From (ii), Jatin scored more in Economics than in Chemistry.

From (ii) and (vi), the marks of Jatin in Zoology and Physics are less than that in Chemistry.

Chemistry cannot be represented by X6 since Economics must be represented by a pie whose value is higher than X6 and no such pies are present in the chart.

Chemistry cannot be represented by X4 because it is the smallest pie in the chart.



If Chemistry is represented by either X2 or X3, then Zoology and Physics cannot be represented by pies which have a lower value than Chemistry. Hence, Chemistry must be represented by X5. Economics must be represented by X6.

Zoology can be represented by X2/X3 or by X4.

Let Zoology be represented by X2/X3.

If Jatin scored $2x$ in Maths, his score in Zoology and one other subject must be x .

Jatin's score in the subject represented by X4 must be $2x - 55$ (from (v)).

Let y be Jatin's score in Economics represented by X6. Jatin's score in Chemistry must be $(y - 15)$ (from (ii)).

His score in Zoology must be $(y - 45)$ (from (vi)).

His score in Maths must be $2y - 90$ (since he scored twice in Maths)

Therefore, $2y - 90 = 2x$; or $y - x = 45$.

Jatin's score in the subject represented by X4 will be $2y - 145$ (since $x = y - 45$)

The sum of the scores of Jatin in the subjects represented by X4, X5 and X6 must be $4x$ (since the sum of scores represented by X1, X2, and X3, which form half the pie, is $4x$)

Therefore, $y + y - 15 + 2y - 145 = 4x$

Or, $4y - 4x = 160$

Or, $y - x = 40$

The two equations above are contradictory.

Hence, Zoology cannot be represented by X2/X3 and must be represented by X4.

Botany and Physics must be represented by X2 and X3 in any order.

In this case, the marks scored in Zoology will be $(y - 45)$.

In Maths, Jatin would have scored $2x$ which must be the same as $y - 45 + 55 = y + 10$.

Hence, $2x = y + 10$, or, $2x - y = 10$.

The marks scored by Jatin in Economics, Chemistry, and Zoology will be (y) , $(y - 15)$, and $(y - 45)$ respectively.

The sum of these three must be equal to $4x$.

Therefore, $3y - 60 = 4x$, or, $-4x + 3y = 60$

Solving the two equations, we get, $y = 80$ and $x = 45$.

Hence, the marks scored in Maths, Physics, Botany, Zoology, Chemistry, and Economics are 90, 45, 45, 35, 65, and 80 respectively.

The total marks scored by Jatin = $8x = 360$.

Video Solution:



Q12. Text Solution:

The marks scored in Maths, Physics, Botany, Zoology, Chemistry, and Economics are 90, 45, 45, 35, 65, and 80 respectively.

The angle subtended by the pie representing Chemistry is 65°

Video Solution:



Q13. Text Solution:

The marks scored in Maths, Physics, Botany, Zoology, Chemistry, and Economics are 90, 45, 45, 35, 65, and 80 respectively.

Zoology is represented by X4.

Video Solution:



Q14. Text Solution:

The marks scored in Maths, Physics, Botany, Zoology, Chemistry, and Economics are 90, 45, 45, 35, 65, and 80 respectively.



Hence marks scored in Botany = 45

Video Solution:



Q15. Text Solution:

The marks scored in Maths (X1), Physics (X2/X3), Botany (X2/X3), Zoology (X4), Chemistry (X5), and Economics (X6) are 90, 45, 45, 35, 65, and 80 respectively.

Hence chemistry is definitely adjacent to zoology.

Video Solution:



Q16. Text Solution:

With the given condition,

The ages of ten people in 2021 = 25, 26, 27, 28, 29, 30, 31, 32, 33, and 34.

The total age = 295

The ten people can be divided into 4 groups in 2 ways i.e., $4 + 2 + 2 + 2$ and $3 + 3 + 2 + 2$.

Looking at the averages of all the groups it is clear that the ten people in 2021 are divided into $4 + 2 + 2 + 2$.

For 2021:

For group P = The age must be 25 and 26.

Similarly, for Q = 27 and 28;

For R = 29 and 30

For S = 31, 32, 33, and 34.

The table below shows the age number in 2021 of those people in all the groups:

Group	2021
P	25, 26
Q	27, 28

R	29, 30
S	31, 32, 33, 34

The ages of the above people will increase by 1.

So, the ages of ten people in 2022 = 26, 27, 28, 29, 30, 31, 32, 33, 34, and 35.

The total age = 305.

Looking at the averages of all the groups it is clear that the ten people;

The year 2022 is divided into $3 + 3 + 2 + 2$.

For 2022:

There must be 2 people in groups Q and S, also 3 people in groups P and R.

For group P (must not be 26 and 27) = The age must be 28, 29, and 31.

Similarly, for Q (must not be 28 and 29) = 26 and 35;

For R (must not be 30 and 31) = 32, 33 and 34.

For S (must not be 32, 33, 34, and 35) = 27 and 30.

The table below shows the age number in 2022 of those people in all the groups:

Group	2022
P	28, 29, 31
Q	26, 35
R	32, 33, 34
S	27, 30

The ages of the above people will increase by 1.

The ages of ten people in 2023 = 27, 28, 29, 30, 31, 32, 33, 34, 35, and 36.

The total age = 315.

For 2023:

Looking at the averages of all the groups it is clear that the ten people;

The year 2022 is divided into $4 + 4 + 2 + 2$.

It is possible only if, $(34.5 \times 4) + (29.5 \times 2) + (28 \times 2) + (31 \times 2) = 138 + 59 + 56 + 62 = 315$

So, groups P, Q, R, and S have 4, 2, 2, and 2 people respectively.

By considering the numbers of the ages of two earlier years the ages:

For group P (must not be 28, 29, 30, and 32) =

The age must be 33, 34, 35, and 36.



Similarly, for Q (must not be 29, 30, 27, and 36) = 28 and 31;

For R (must not be 31, 32, 33, 34, and 35) = 27 and 29.

For S (must not be 33, 34, 35, 36, 28, and 31) = 30 and 32.

The table below shows the age number in 2022 of those people in all the groups:

Group	2023
P	33, 34, 35, 36
Q	28, 31
R	27, 29
S	30, 32

The following table is the final one:

Group	2021	2022	2023
P	25, 26	28, 29, 31	33, 34, 35, 36
Q	27, 28	26, 35	28, 31
R	29, 30	32, 33, 34	27, 29
S	31, 32, 33, 34	27, 30	30, 32

The age of the eldest person in Group Q on January 1, 2021, is 28 years.

Video Solution:



Q17. Text Solution:

Group	2021	2022	2023
P	25, 26	28, 29, 31	33, 34, 35, 36
Q	27, 28	26, 35	28, 31
R	29, 30	32, 33, 34	27, 29
S	31, 32, 33, 34	27, 30	30, 32

The age will increase by 1 from 2021 to 2022.

The ages of people in group P in 2021 = 25 and 26.

So, the age of the same person will be 26 and 27 in 2022.

The ages of people in group S in 2022 = 27 and 30.

Hence, one person is common.

Video Solution:



Q18. Text Solution:

Group	2021	2022	2023
P	25, 26	28, 29, 31	33, 34, 35, 36
Q	27, 28	26, 35	28, 31
R	29, 30	32, 33, 34	27, 29
S	31, 32, 33, 34	27, 30	30, 32

The required sum = 26 + 35 = 61.

Video Solution:



Q19. Text Solution:

Group	2021	2022	2023
P	25, 26	28, 29, 31	33, 34, 35, 36
Q	27, 28	26, 35	28, 31
R	29, 30	32, 33, 34	27, 29
S	31, 32, 33, 34	27, 30	30, 32

Group P showed the highest increase in total age from 2021 to 2023.

Video Solution:



Q20. Text Solution:

Group	2021	2022	2023
P	25, 26	28, 29, 31	33, 34, 35, 36
Q	27, 28	26, 35	28, 31
R	29, 30	32, 33, 34	27, 29
S	31, 32, 33, 34	27, 30	30, 32

33 was the age of the youngest person in Group P on January 1, 2023.

Video Solution:**Q21. Text Solution:**

Let the number of questions correctly answered by Dayal in Singing be 'x'.

Dayal's number of correct responses in Poetry will be $(15 - x)$, which will also be Benny's number of correct responses in History as given.

Rekha's number of correct responses in singing will be $(19 - x)$ given singing total correct is 34, and thus, Rekha's number of correct responses in history is $(x - 8)$, given her total of 18 corrects.

So, a table with correct questions in each section can be prepared with the said information as below:

	Singing	Poetry	History	Total Corrects	Total Score
Benny	8		$15 - x$		63
Dayal	x	$15 - x$	6	21	
Prasoon	7		9		
Rekha	$19 - x$	7	$x - 8$	18	
Total	34	36	22	92	

Given that each participant has got at least 2 questions correct for each section, ' $x \geq 10$ ', using Rekha's value for history, and ' $x \leq 13$ ' using Benny's value for history.

Since we are given the total score of Benny = 63

We will have to use it to find the final table values.

Let Benny's score in Poetry = y;

For Benny's total score:

Case 1: When $x = 10$,

$$= 16 + y + 5 = 63$$

$y = 42$, which means Benny must have answered more than 10 questions correctly, which means she must have scored 3 marks for each correct answer.

So, Benny's correct answers in Poetry = 14, if that is the case, she must have scored bonus marks also, which is not possible.

Case 2 : When $x = 11$,

$$20 + y = 63$$

$y = 43$, which means Benny must have answered more than 10 questions correctly, which means she must have scored 3 marks for each correct answer.

This case is possible if she must have scored bonus marks, that 13 questions are answered correctly in poetry.

This case is possible.

Case 3: When $x = 12$, & Case 4: When $x = 13$, we see that the parameters do not satisfy given conditions, hence are not possible.

Therefore, only case 2 is valid, i.e., $x = 11$, by which calculating the complete table as below:

	Singing	Poetry	History	Total Corrects	Total Score
Benny	8	13	4	25	63
Dayal	11	4	6	21	49
Prasoon	7	12	9	28	68
Rekha	8	7	3	18	33
Total	34	36	22	92	

For all four students, we can uniquely determine the number of correct questions attempted by them.

Hence answer is 4.

Video Solution:



Q22. Text Solution:

	Singing	Poetry	History	Total Corrects	Total Score
Benny	8	13	4	25	63
Dayal	11	4	6	21	49
Prasoon	7	12	9	28	68
Rekha	8	7	3	18	33
Total	34	36	22	92	

Benny answered a total of 25 questions correctly.

Video Solution:



Q23. Text Solution:

	Singing	Poetry	History	Total Corrects	Total Score
Benny	8	13	4	25	63
Dayal	11	4	6	21	49
Prasoon	7	12	9	28	68
Rekha	8	7	3	18	33
Total	34	36	22	92	

Prasoon scored highest and hence would have been stood at 1st position.

Video Solution:



Q24. Text Solution:

	Singing	Poetry	History	Total Corrects	Total Score
Benny	8	13	4	25	63
Dayal	11	4	6	21	49
Prasoon	7	12	9	28	68
Rekha	8	7	3	18	33
Total	34	36	22	92	

A. His total score lies between 50 to 55: Incorrect

B. He answered six questions correctly in Poetry: Incorrect

C. He answered the second-lowest number of correct questions in Poetry: Incorrect

D. He answered the highest number of correct questions in Singing among all four students: Correct

Video Solution:



Q25. Text Solution:

	Singing	Poetry	History	Total Corrects	Total Score
Benny	8	13	4	25	63
Dayal	11	4	6	21	49
Prasoon	7	12	9	28	68
Rekha	8	7	3	18	33
Total	34	36	22	92	

A. He got 12 questions correct in Poetry: Correct.

B. His total score is the second lowest: Incorrect.

C. His total score is the arithmetic mean of the total scores of Dayal and Benny: Incorrect

D. His total score is the second highest: Incorrect

Video Solution:



**Q26. Text Solution:**

The variable cost of the biryani = (cost of chicken + rice cost + egg's cost + processing cost + spice cost)

So, the variable cost per biryani is

$$(150 \times 0.2 + 60 \times 0.25 + \frac{60}{12} +$$

$$60 + 30) \text{INR} = (30 + 15 + 5 + 90) \text{INR} = 140 \text{INR}$$

The fixed cost per month (assuming one cook) is (500

$$30 + 15,000 + 10,000) \text{INR} = 40,000 \text{INR}$$

Let us assume that a minimum of n number of biryani plates needs to be sold in a month to avert the losses.

So,

$$300n \geq 140n + 40,000$$

$$\Rightarrow n \geq 40,000/160$$

$$\Rightarrow n \geq 250$$

Thus, the number of biryani needed to be sold to make sure no loss is 250 plates.

As everyday the average demand is the same and $\frac{250}{30}$ is not an integer, so the answer will be 270 plates.

The answer is option C.

Video Solution:**Q27. Text Solution:**

The variable cost of the biryani = (cost of chicken + rice cost + egg's cost + processing cost + spice cost)

So, the variable cost per biryani is

$$(150 \times 0.2 + 60 \times 0.25 + \frac{60}{12} +$$

$$60 + 30) \text{INR} = (30 + 15 + 5 + 90) \text{INR} = 140 \text{INR}$$

The fixed cost per month (assuming one cook) is (500

$$30 + 15,000 + 10,000) \text{INR} = 40,000 \text{INR}$$

Let us assume that a minimum of n number of biryani plates needs to be sold in a month to avert the losses.

So,

$$240n \geq 140n + 40,000$$

$$\Rightarrow 100n \geq 40,000$$

$$\Rightarrow n \geq 400$$

Now, one cook can not handle pressure of more than $30 \times 10 = 300$ biryani. So, one more cook is needed for all the days as the daily average biryani requirement is more than 10.

So, considering 2 cooks we get–

$$240n \geq 140n + 40,000 + 15,000$$

$$\Rightarrow 100n \geq 55,000$$

$$\Rightarrow n \geq 550$$

$$\text{The per day demand is } \frac{550}{30} = 18.33$$

As the demand per day is the same and it cannot be a fraction, so the minimum number of daily demands is 19.

$$\text{Thus, the minimum of plates of biryani is } 19 \times 30 = 570$$

The answer is option C.

Video Solution:**Q28. Text Solution:**

In a 30 day month, a total of 300 plates can be made by 1 cook. So, to have 550 plates we need at least 2 cooks.

To minimise the cost, we need to make sure that the second cook gets fully utilised like the first cook.

So, the second cook will be used for only

$$\frac{550-300}{10} \text{ days} = 25 \text{ days}$$



So, the minimum possible cook cost is
 $500 \times (30 + 25) = 27500 \text{ INR}$
 Total cost = $(27,500 + 25,000) \text{ INR} = 52,500 \text{ INR}$
 So, total revenue less the variable cost is
 $(240 - 140) \times 550 = 55,000 \text{ INR}$
 Maximum possible profit is $(55000 - 52500) \text{ INR}$
 $= 2500 \text{ INR}$
 The answer is option A.

Video Solution:



Q29. Text Solution:

Total number of biryanis produced per day = $\frac{900}{30}$
 $= 30 \text{ plates}$
 So, 3 cooks are needed.
 Hence, the fixed cost is $15000 \times 3 + 15000 + 10000 = 70,000 \text{ INR}$
 Revenue after removing the per-plate cost of chicken biryani,
 $= 900 \times (200 - 140) \text{ INR} = 54000 \text{ INR}$
 So, the loss is $(70,000 - 54,000) \text{ INR} = 16,000 \text{ INR}$
 The answer is option B.

Video Solution:



Q30. Text Solution:

Total number of biryanis produced per day = $\frac{900}{30}$
 $= 30 \text{ plates}$
 So, 3 cooks are needed.
 Hence, the fixed cost is $15000 \times 3 + 15000 + 10000 = 70,000 \text{ INR}$
 The revenue less variable cost for these 900 plates

$= 900 \times (200 - 140 + 30) \text{ INR} = 81,000 \text{ INR}$
 So, the profit is $(81,000 - 70,000) \text{ INR} = 11,000 \text{ INR}$
 The answer is option B.

Video Solution:

